

Lesson 3A Reference: Output, Repetition & Quotes

Everything taught in class today, on one page. Keep it — you'll want it in October.

You built: the Tournament Banner (`lesson3A_banner_starter.py`)

Quick translation table

JavaScript	Python
<code>console.log("Hi")</code>	<code>print("Hi")</code>
<code>console.log(a, b)</code>	<code>print(a, b)</code>
<code>"".repeat(20)</code>	<code>"*" * 20</code>
<code>// comment</code>	<code># comment</code>
<code>let player = "Nova"</code>	<code>player = "Nova"</code>
<code>studentName</code>	<code>student_name</code>
<code>;</code> at end of line	omit it

print()

```
print("Hello, World!")           # Hello, World!
print("Welcome", "to", "Python") # Welcome to Python
print("Sum is", 7 + 3)           # Sum is 10
```

Two things to notice:

- **Commas add a space for you.** You don't type it.
- **Math happens first**, then gets printed. `7 + 3` becomes `10` before it appears.

No semicolons. They're legal but PEP 8 forbids them, and style is part of your grade. Ruff will underline them.

sep — what goes BETWEEN

```
print("a", "b", "c")           # a b c      (default: one space)
print("a", "b", "c", sep="-")  # a-b-c
print("a", "b", "c", sep="")   # abc
print("a", "b", "c", sep=" > ") # a > b > c
```

JavaScript has no equivalent — you'd build the string with `.join()` first.

⚠ The trap: sep separates EVERYTHING

```
print("Rounds:", 13, 9, sep=" > ") # Rounds: > 13 > 9
```

The label is an item too, so it collects a separator of its own. Two ways around it:

```
# Fix 1 — print the label separately, held open with end=" "
print("Rounds:", end=" ")
print(13, 9, sep=" > ")           # Rounds: 13 > 9

# Fix 2 — fold the label into the first item
print("Rounds: 13", 9, sep=" > ") # Rounds: 13 > 9
```

Both work. Fix 1 is clearer when the label is long; Fix 2 is shorter.

end — what goes AFTER

Every `print()` finishes with a newline unless you say otherwise.

```
print("Loading")           # ends the line (the default)
print("Done")
# Loading
# Done

print("Loading", end="")   # ends with nothing
print("...done")
# Loading...done
```

Use `end=""` when you want several prints to land on one line. `console.log` can't do this at all.

Repeating text with `*`

```
print("-" * 40)           # 40 dashes
print("Hi! " * 3)         # Hi! Hi! Hi! (the space repeats too)
print(20 * "*")           # order doesn't matter
```

A string times a **number** repeats it.

The trap: never `string × string`

```
print("*" * "20")
# TypeError: can't multiply sequence by non-int of type 'str'
```

This matters more than it looks. `input()` always hands you a *string*, so you'll meet this error again in Lesson 5.

Building a box

The border must match the width of the middle line:

```
print("=" * 10)
print("= PYTHON =")  # this line is 10 characters
print("=" * 10)
```

Centering — calculate, don't count

```
WIDTH = 40
title = "BERGEN TECH OPEN"

padding = (WIDTH - len(title)) // 2    # (40 - 16) // 2 = 12
print(" " * padding + title)
```

`len(title)` gives the string's length. `//` divides and throws away the remainder, so you get a whole number of spaces.

Quotes inside strings

⚠ Why this breaks

```
print('Display 'hi' in quotes')    # SyntaxError
```

Python thinks the string ends at the second `'`. Everything after it is nonsense to the interpreter.

Fix 1 — switch the outer quotes (cleanest)

```
print('She said "nice play"')      # double inside single
print("It's Bergen Tech's best team") # single inside double
```

Reach for this first.

Fix 2 — escape with a backslash

```
print('Display \'hi\' in quotes')
print("Display \"hi\" in quotes")
```

Works, but harder to read. The `\` tells Python "this quote is data, not the end of the string."

When both quote types appear

Switching the outer quote can't save you, because both are inside:

```
print("He said \"that's mine\"")
```

Escaping works. There's also a cleaner way using triple quotes — that's Lesson 3B.

Apostrophes are the usual culprit: *O'Brien, don't, it's, you're*.

Common errors

Error or symptom	Cause	Fix
<code>SyntaxError</code> on a quoted line	Same quote type inside and outside	Switch the outer quotes, or escape
<code>TypeError: can't multiply sequence by non-int</code>	Multiplied a string by a string	The right side must be a number
Extra separator before your first value	<code>sep</code> also separates the label	Use <code>end=" "</code> , or fold the label in
<code>NameError: name 'x' is not defined</code>	Variable never created, or a typo	Check spelling and that it's assigned above
Border doesn't line up	Middle line is a different width	Count characters, or calculate with <code>len()</code>
Your changes don't appear	File wasn't saved	Ctrl/Cmd + S, then run again

Checklist before you push

- ☐ Header comment at the top: your name, assignment, date
- ☐ No semicolons anywhere
- ☐ `snake_case` variable names, not `camelCase`
- ☐ The file **runs** without errors
- ☐ Output matches the target exactly, including spaces

- ☐ Padding and widths are **calculated**, not counted by hand
 - ☐ Committed **and pushed** — check github.com to confirm
-

Next lesson: f-strings, escape sequences, and triple quotes.